

AI-Driven Job—Skill—Curriculum Alignment Engine

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MOTIVATION

Students ask: “What jobs am I qualified for with my current skills?”

- Job postings are difficult for students to interpret
- Curriculum language differs from workforce language

Opportunity: connect job postings $\leftrightarrow$ skills $\leftrightarrow$ coursework

VISION

Build an AI engine that:

- Ingests real job postings
- Extracts and normalizes required skills
- Models skills from courses and student's previous experience
- Matches students to jobs (job recommendation)
- Recommends courses to close skill gaps
- Explains results using GenAI

TECHNICAL ARCHITECTURE OVERVIEW

Five core components:

1. Job Posting Ingestion & Skill Extraction
2. Curriculum & Course Skill Modeling
3. Student Skill Profile Construction
4. Matching & Recommendation Engine
5. GenAI Explanation Layer

COMPONENT 1 – Job Posting Ingestion & Skill Extraction

- Collect CS/CE-related job postings (e.g., software, data, cybersecurity, AI roles)
- Utilize NLP / LLMs to extract:
 - Technical skills
 - Tools and technologies
- Compare extraction approaches (keyword vs embeddings vs LLM)
- Normalize skills

COMPONENT 2 – Curriculum & Course Skill Modeling

- Represent each course as a set of skills
- Use:
 - Course Descriptions (from CourseLeaf)
 - Course Learning Outcome (from CourseLeaf)
 - Syllabi (from Instructors (CS: already obtained))
- Encode as course $\rightarrow$ skill vectors
- Lightweight, interpretable representation

COMPONENT 3 – Student Skill Profile Construction

- Build student skill profile from
 - Completed courses
 - Optional self-reported skills
- Aggregate skills into a student skill vector

COMPONENT 4 – Matching & Recommendation Engine

- Match student skills to job skill requirements
- Rank job postings by alignment
- Identify skill gaps and Recommend courses to close gaps

COMPONENT 5 – GenAI Explanation Layer

- Explain:
 - Why jobs are recommended
 - Which skills the student already has
 - Which skills are missing
- Recommend courses with clear rationale

PHASE 1 SCOPE – (6 Months, CS Only)

Includes:

- Job posting dataset
- Skill extraction pipeline
- Course and student skill models
- Matching engine
- GenAI explanation UI

Excludes:

- Degree pathway optimization
- Multi-major support

TEAM & FEASIBILITY

Effort estimate: 2 MS students

Timeline: 6 months

Roles:

- Research-focused student: NLP + matching
- Second student: UI + GenAI integration